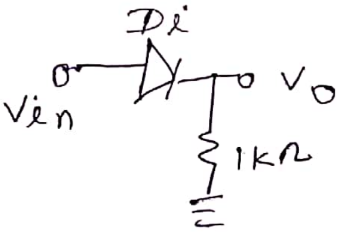
Assignment 2

1. i/p = 5 sin ωt

(a)



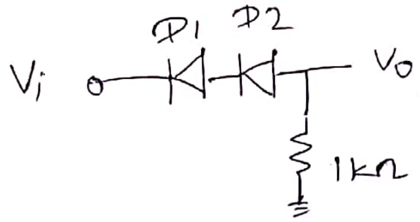
Vin > 0 Diode on

Vo = Vin = 5V

Vin < 0 Diod off

Vo = 0 V

(b)



vin > 0

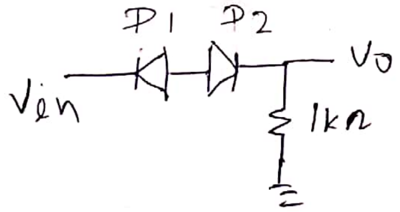
D1, D2 OFF

Vo = 0

~~vi>0~~ vi < 0 D1 & D2 on

Vo = Vi = 5V

(c)



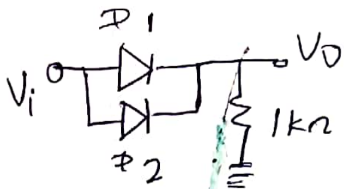
Vin > 0 D1 off

VO = 0

vi < 0 D1 on. D2 = OFF

Vo = 0

(d)



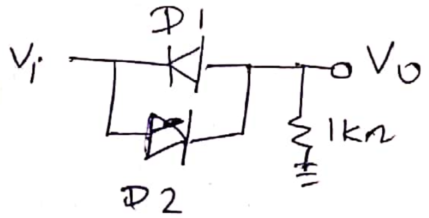
Vi > 0 D1, D2 on

Vo = Vi = 5V

Vi < 0 D1, D2 OFF

Vo = 0

(e)



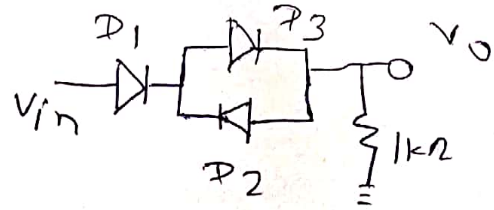
Vi > 0 D2 on D1 OFF

Vo = Vin = 5V

Vi < 0 D1 on D2 OFF

Vo = Vin = 5V

(f)



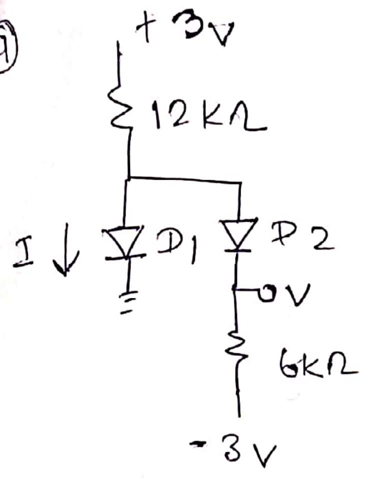
Vi > 0 D1 on D3 on
D2 OFF

Vo = Vin = 5V

Vin < 0 Di OFF

Vo = 0

(a)



1st D2 will on

D1 will OFF

$$\frac{3-(-3)}{18\,k\Omega} = \frac{6}{18}\,mA = \frac{1}{3}\,mA$$

then $V - 6k\Omega \times \frac{1}{3}mA = -3$

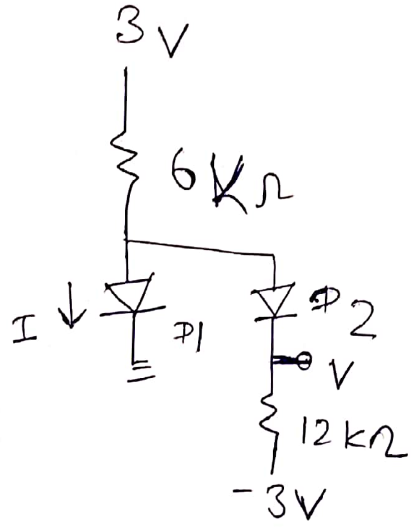
$V - 2 = -3$

$V = -3+2 = -1V$

so D1 will off

$I = 0 \quad V = -1V$

(b)



Ist D2 will on

$I = \frac{1}{3}mA$

$V - 12k\Omega \times \frac{1}{3}mA = -3$

$V - 4 = -3$

$V = 4-3 = 1V$

D1 will on

$$\frac{0-3}{6k\Omega} + \frac{0-(-3)}{12k\Omega} + I = 0$$

$$-\frac{1}{2}mA + \frac{1}{4}mA + I = 0$$

$$+\frac{1}{4}mA = +I$$

$$\boxed{I = \frac{1}{4}mA \quad V = 0}$$

$\ln 10000 = \frac{V}{0.026} \Rightarrow V = 0.239\,V$

(b) $I = I_s\, e^{\frac{0.7}{0.026}}$

$I = 4.9 \times 10^{11} \times I_s$

(4) $T = 400k$ $\quad A = 100\mu m^2$ $\quad L_n = 20\mu m$ $\quad L_p = 30\mu m$

$N_A = 2\times10^{16}\, cm^{-3}$ $\quad N_D = 4\times10^{16}\, cm^{-3}$

$$I_s = \frac{eA\, D_n\, p_n}{L_n} + \frac{eA\, D_p\, n_p}{L_p}$$

$D_n = \frac{KT}{q}\mu_n$ $\quad D_p = \frac{KT}{q}\mu_p$

$\mu_n = 1300\, cm^2/V.sec$ $\quad \mu_p = 500\, cm^2/Vsec$

$K = 1.38 \times 10^{-23}$

$$p_n = \frac{n_i^2}{n_n} = \frac{2.25\times10^{20}}{4\times10^{16}} = 5625.0$$

$$n_p = \frac{2.25\times10^{20}}{2\times10^{16}} = \frac{2.25\times10000}{2} = 11250$$

$$D_n = \frac{1.38\times10^{-23}\times400}{1.6\times10^{-19}}\times1300 = 0.0345\times1300 = 44.85$$

$$D_p = \frac{1.38\times10^{-23}\times400\times500}{1.6\times10^{-19}} = 17.25$$

$$I_s = eA\left(\frac{44.85\times\cancel{5625}^{\,11250}}{20\times10^{-6}} + \frac{17.25\times\cancel{11250}^{\,5625}}{30\times10^{-6}}\right)$$

$$= 1.6\times10^{-19}\times100\times10^{-6}\left(\cancel{12614.06}^{\,25228.125\times10^{6}} + \cancel{6468.75}^{\,3234.375\times10^{6}}\right)$$

$I_s = \cancel{3.05\times10^{-19}A}$ $\quad 28462.5\times100\times10^{-19}$

$$\boxed{I_s = 2.8\times10^{-13}}$$

(5) $C_J = \frac{A}{2}\left[\frac{2q\varepsilon}{(V_0 - V)} \frac{N_A N_D}{N_A + N_d}\right]^{1/2}$

$V_0 = 0.026 \ln \frac{n_n}{p_n} = 0.026 \ln \frac{4\times10^{16}}{p_n}$

$p_n = \frac{n_i^2}{n_n} = \frac{2.25\times10^{20}}{4\times10^{16}} = \frac{2.25\times10000}{4} = 5625$

$V_0 = 0.026 \ln \frac{4\times10^{16}}{5625} = 0.769\,V$

when $V = 0$

$C_J = \frac{100\times10^{-6}}{2}\left[\frac{2\times1.6\times10^{-19}\times8.85\times10^{-14}}{0.769} \times \frac{8\times10^{32}}{2\times10^{16}+4\times10^{16}}\right]^{1/2}$

$= 50\times10^{-6}\left[\frac{2\times1.6\times8.85}{0.769}\times10^{-33} \times \frac{8\times10^{16}}{6}\right]$

$= 50\times10^{-6}\left[\frac{2\times1.6\times8.85\times8}{0.769\times6}\times10^{-17}\right]^{1/2}$

$= \cancel{\frac{50\times2\times1.6\times8.85\times8\times10^{-23}}{0.769\times6}}$

$\cancel{= 2455\times10^{-23}}$

$= 50\times10^{-6}\times2.21\times10^{-8}$

$= 11.05\times10^{-15}\,F$

when $V = -1V$

$C_J = 50\times10^{-6}\left[\frac{2\times1.6\times8.85\times8\times10^{-17}}{1.769\times6}\right]^{1/2}$

$= 50\times10^{-6}\times1.46\times10^{-8} = 73.05\times10^{-14}$

$= 7.305\times10^{-15}\,F$

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8 $N_A = 10^{17}\ cm^{-3}$ $N_D = 10^{15}\ cm^{-3}$

-o Vout

$$V_{bi} = \frac{K.T}{q} \ln\left(\frac{N_A N_D}{N_i^2}\right)$$

$$= 0.026 \ln\left(\frac{10^{17} \times 10^{15}}{2.25 \times 10^{20}}\right)$$

$$= 0.026 \ln\left(\frac{10^{12}}{2.25}\right)$$

$$\boxed{V_{bi} = 0.697\,V}$$

$$W = \left[\frac{2\varepsilon V_0}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$$

$$= \left[\frac{2 \times 8.85 \times 10^{-14} \times 0.697}{1.6 \times 10^{-19}}\left(\frac{10^{17} + 10^{15}}{10^{32}}\right)\right]^{1/2}$$

$$= \left[\frac{2 \times 8.85 \times 0.697 \times 10^{-14+19-15}(1+0.01)}{1.6}\right]$$

$$w = \left(\frac{2 \times 8.85 \times 0.697 \times 10^{-10} \times 1.01}{1.6}\right)^{1/2}$$

$$= 2.79 \times 10^{-7}\ cm$$

$$= 0.279\ \mu m$$

$$V_0 = -\frac{1}{2} E_0 W \Rightarrow E_0 = -\frac{2 V_0}{W}$$

$$E_0 = -\frac{2 \times 0.697}{0.279 \times 10^{-6}}$$

$$\boxed{E_0 = -4.996 \times 10^{6}\ V/m}$$

$N_D = 10^{16}\ cm^{-3}$

$A = 2 \times 10^{-3}\ cm^2$

$N_A = 4 \times 10^{18}\ cm^{-3}$

$V_0 \quad x_n \quad x_p \quad E_m$

$$V_0 = \frac{KT}{e} \ln \left(\frac{10^{16} \times 4 \times 10^{18}}{2.25 \times 10^{20}} \right)$$

$$V_0 = 0.026 \ln \frac{10^{14} \times 4}{2.25}$$

$$\boxed{V_0 = 0.85\ V}$$

$$W = \left[\frac{2 \varepsilon V_0}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}$$

$$= \left(\frac{2 \times 8.85 \times 10^{-14} \times 0.85}{1.6 \times 10^{-19}} \left(\frac{10^{16} + 4 \times 10^{18}}{4 \times 10^{34}} \right) \right)^{1/2}$$

$$\boxed{W = 10^{-5} \times 0.977\ cm = 0.997 \times 10^{-5}\ cm}$$

$x_n + x_p = 0.97 \times 10^{-7}\ m$

$x_n N_d = x_p N_A$

$x_n = x_p \times 400$

$400 x_p + x_p = 0.97 \times 10^{-7}$

$x_p (401) = 0.97 \times 10^{-7}$

$$x_p = \frac{0.97}{401} \times 10^{-7}$$

$$\boxed{\begin{array}{l} x_p = 0.0024 \times 10^{-7}\ m \\ x_n = 0.967 \times 10^{-7}\ m \end{array}}$$

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$N_D = 10^{18}\ cm^{-3}$ $N_A = 10^{16}\ cm^{-3}$

$\tau_p = \tau_n = 10^{-6}\ s$ $D_n = 21\ cm^2/s$

$D_p = 10\ cm^2/s$ $A = 1.2 \times 10^{-5}\ cm^2$

Vout

(a) Saturation current at 300K

(b) Calc the forward reverse current at $\pm 0.7V$

$$P_n = \frac{n_i^2}{n_n} = \frac{2.25 \times 10^{20}\ cm^{-6}}{10^{18}\ cm^{-3}} = 2.25 \times 10^{2}\ cm^{-3}$$

$$n_p = \frac{n_i^2}{P_p} = \frac{2.25 \times 10^{20}\ cm^{-6}}{10^{16}\ cm^{-3}} = 2.25 \times 10^{4}\ cm^{-3}$$

$$L_n = (D_n \tau_n)^{1/2} = (21 \times 10^{-6})^{1/2} = 4.58 \times 10^{-3}\ cm$$

$$L_p = (D_p \tau_p)^{1/2}$$

$$= (10 \times 10^{-6})^{1/2} = 3.16 \times 10^{-3}\ cm$$

$$I_0 = \frac{eA D_p P_n}{L_p} + \frac{eA D_n n_p}{L_n}$$

$$= 1.6 \times 10^{-19} \times 1.2 \times 10^{-5} \left(\frac{10 \times 2.25 \times 10^{2}}{3.16 \times 10^{-3}} + \frac{21 \times 2.25 \times 10^{4}}{4.58 \times 10^{-3}} \right)$$

$$= 1.6 \times 1.2 \times 10^{-24} \left(\frac{22.5 \times 10^{5}}{3.16} + \frac{21 \times 2.25 \times 10^{7}}{4.58} \right)$$

$$= 1.6 \times 1.2 \times 10^{-24} \times 10^{7} \left(\frac{0.225}{3.16} + \frac{21 \times 2.25}{4.58} \right)$$

$$= 1.6 \times 1.2 \times 10^{-17} (10.31 + 0.071)$$

$$= 19.93 \times 10^{-17}$$

$$\boxed{I_0 = 1.936 \times 10^{-16}}\ A$$

$$I = 1.936 \times 10^{-16} \left(e^{\frac{0.7}{0.026}} \right)$$

$$= 9.56 \times 10^{-5}\ A$$

$I = I_s e^{\frac{V}{nV_T}}$

$I_s = \frac{eA D_p p_n}{L_p} + \frac{eAD_n n_p}{L_n}$

$P_n = \frac{2.25 \times 10^{20}}{4 \times 10^{16}}$

$= 0.5625 \times 10^4$

$D_n = \frac{KT}{q} \mu_n \qquad D_p = \frac{KT}{q} \mu_p$

$n_p = \frac{2.25 \times 10^{20}}{2 \times 10^{16}}$

$= 1.125 \times 10^4$

given $\quad I = 2 \times 10^{-17} A \qquad V_R = -1V \qquad A = 100 \mu m^2$

$\mu_n = 1350 cm^2/V\text{-}s \qquad \mu_p = 480 \, cm^2/V\text{-}s$

$N_A = 2 \times 10^{16} cm^{-3} \qquad N_D = 4 \times 10^{16} cm^{-3}$

$I_s = eA \left(\frac{n_p \frac{KT}{q} \mu_n}{L_n} + \frac{\frac{KT}{q} \mu_p P_n}{L_p} \right)$

$= \cancel{e} A \times \frac{KT}{\cancel{q}} \left(\frac{n_p \mu_n}{L_n} + \frac{\mu_p P_n}{L_p} \right)$

$I_s = 100 \times 10^{-6} \times 1.38 \times 10^{-23} \, T \left(\frac{1350 \times 1.125 \times 10^4}{L_n} + \frac{480 \times 0.5625 \times 10^4}{L_p} \right.$

$L_n = \sqrt{D_n \tau_n} = \sqrt{\frac{KT}{q} \mu_n \times}$

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$N_D = 10^{16}\ cm^{-3}$

$V = 0.8\ V$

$$n_n \times p_n = n_i^2$$

$$p_n = \frac{2.25 \times 10^{20}}{10^{16}}$$

$$= 2.25 \times 10^{4}\ cm^{-3}$$